1 Copy of Question 1 (6 Points)

Answer the following questions:

a) What is the acting rationally? (2 points)
b) What is thinking rationally? (2 points)
c) How can an agent act rationally without thinking rational? (2 points)
Skriv in ditt svar här

Teckenf…

Ord: 0

Totalpoäng: 6

2 Copy of Question 2 (13 Points)

1) An agent might be placed into an environment that is fully observable or partially observable, deterministic or stochastic, single-player or multi-player, episodic or sequential. The adjectives “partially observable,” “stochastic,” “episodic,” and “single-agent” are used to describe different types of gaps in the agent’s knowledge. Define each of these four adjectives: specifically, what type of knowledge about the environment is lacking, in each of these four cases? (8 points)

2) Explain Utility based agents. (5 points)

Skriv in ditt svar här

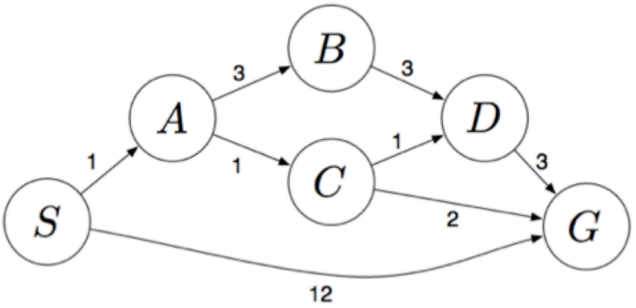
Teckenf…

Ord: 0

Totalpoäng: 13

3 Copy of Question 3 (10 Points)

Consider the search problem with the following state space:



S denotes the start state, G denotes the goal state, and step costs are written next to each arc. Assume that ties are broken alphabetically (i.e., if there are two states with equal priority on the frontier, the state that comes first alphabetically should be visited first).

a) What path would BFS return for this problem? Explain in details. (3 points)

b) What path would DFS return for this problem? Explain in details. (3 points)

c) In general, discuss the relative strengths and weaknesses of breadth-first search (BFS) vs. depth-first search (DFS) for AI problems. (4 points)

	BFS	**DFS**
Strength		
Weakness		

Skriv in ditt svar här

Teckenf...

Ord: 0

Totalpoäng: 10

4 Copy of Question 4 (18 points)

1) Convert the following sentences to first-order logic:

a) All dogs have tails (2 points)

b) John is a student (2 points)

c) Every dog chases some rabbit (4 points)

d) Everyone who loves Santa loves any reindeer (4 points)

e) Anyone who wears maroon-and-white shirts is an Aggie (6 points)

Skriv in ditt svar här

Teckenf…

Ord: 0

Totalpoäng: 18

5 Copy of Question 5 (9 points)

Write down three Ethical aspects of Artificial Intelligence and explain each of these aspects with examples. (9 points)

Skriv in ditt svar här

Teckenf…

Ord: 0

Totalpoäng: 9

6 Copy of Question 6 (22 Points)

Suppose a genetic algorithm uses chromosomes of the form x = abcdefgh with a fixed length of eight genes. Each gene can be any digit between 0 and 9. Let the fitness of individual x be calculated as:

$$f(x) = (a + b) - (c + d) + (e + f) - (g + h) ,$$

and let the initial population consist of four individuals with the following chromosomes:

x1 = 6 5 4 1 3 5 3 2
x2 = 8 7 1 2 6 6 0 1
x3 = 2 3 9 2 1 2 8 5
x4 = 4 1 8 5 2 0 9 4

a) Evaluate the fitness of each individual, showing all your workings, and arrange them in order with the fittest first and the least fit last. (2 points)

b) Perform the following crossover operations: (6 points)

i) Cross the fittest two individuals using one–point crossover at the middle point.
ii) Cross the second and third fittest individuals using a two–point crossover (points b and f).
iii) Apply a mutation operator to the least fittest individual and change one gene value only (select the gene and the new value should be between 0 and 9).

c) Suppose the new population consists of the five offspring individuals received by the crossover and mutation operations in the above question. Evaluate the fitness of the new population, showing all your workings. Has the overall fitness improved? (5 points)

d) By looking at the fitness function and considering that genes can only be digits between 0 and 9 find the chromosome representing the optimal solution (i.e. with the maximum fitness). Find the value of the maximum fitness. (5 points)

e) By looking at the initial population of the algorithm can you say whether it will be able to reach the optimal solution? (4 points)

Skriv in ditt svar här

Teckenf...

Ord: 0

Totalpoäng: 22

7 Copy of Question 7 (16 points)

(16 points) The maze is as follows:

```
##########
#S...#...#
###.#.#.#.#
#....#.#G#
##########
```

You are given a grid-based maze where each cell represents a position. Some cells are blocked (marked as `#`), and some are open (marked as `.`). The goal is to navigate from a **start position** (`S`) to a **goal position** (`G`) using an AI search algorithm.

Constraints:

1. The agent can move **up, down, left, or right** but cannot move diagonally.
2. Blocked cells (`#`) cannot be traversed.
3. The cost of moving between any two adjacent cells is 1.

Tasks:

1. **Representation:** Represent the maze in a way that an AI search algorithm can process (e.g., as a graph or matrix).
2. **Algorithm Application:** Solve the problem using A* Search (A-Star) with Manhattan Distance as the heuristic function.
3. **Solution Path:** Show the solution step by step in details.
4. **Solution Path:** Write down the sequence of moves (e.g., `Right`, `Down`, `Left`, etc.) that leads from `S` to `G` using the chosen algorithm.
5. **Visualization:** Optionally, mark the solution path on the maze by replacing the traversed cells with `*` (excluding `S` and `G`).

Example Solution:

The students should:

- Represent the maze as a graph.
- Demonstrate the working of the chosen algorithm step-by-step (e.g., exploration of nodes or queue updates).
- Provide the final solution path, e.g., `Right, Right, Down, Down, Down`.

Skriv in ditt svar här

Teckenf…

Ord: 0

Totalpoäng: 16

8 Copy of Question 8 (6 points)

What are the differences between Supervised and Unsupervised learning (**write down only 3**). (6 points)

Skriv in ditt svar här

Teckenf…

Ord: 0

Totalpoäng: 6